

- A. Vaswani et al., "Attention Is All You Need," 2017.
- N. Shazeer, "GLU Variants Improve Transformer," 2020.



# Deep dive into Transformer

## 1. Our goals

This project pursues three objectives. First, we aim to deeply understand the Transformer architecture by implementing it from scratch without relying on pre-built libraries. Second, we investigate two variations of the Transformer: **Encoder-Decoder (ED)** and **Decoder-only (D-only)**. Third, we compare performance of different architectural choices—normalization placement, activation functions, and positional encoding.

## 2. Transformer

The Transformer relies on scaled dot-product attention (Fig. 1, left). Multiple attention heads run in parallel (Fig. 1, right), allowing the model to attend to different representation subspaces. Each attention layer is followed by a feed-forward network.

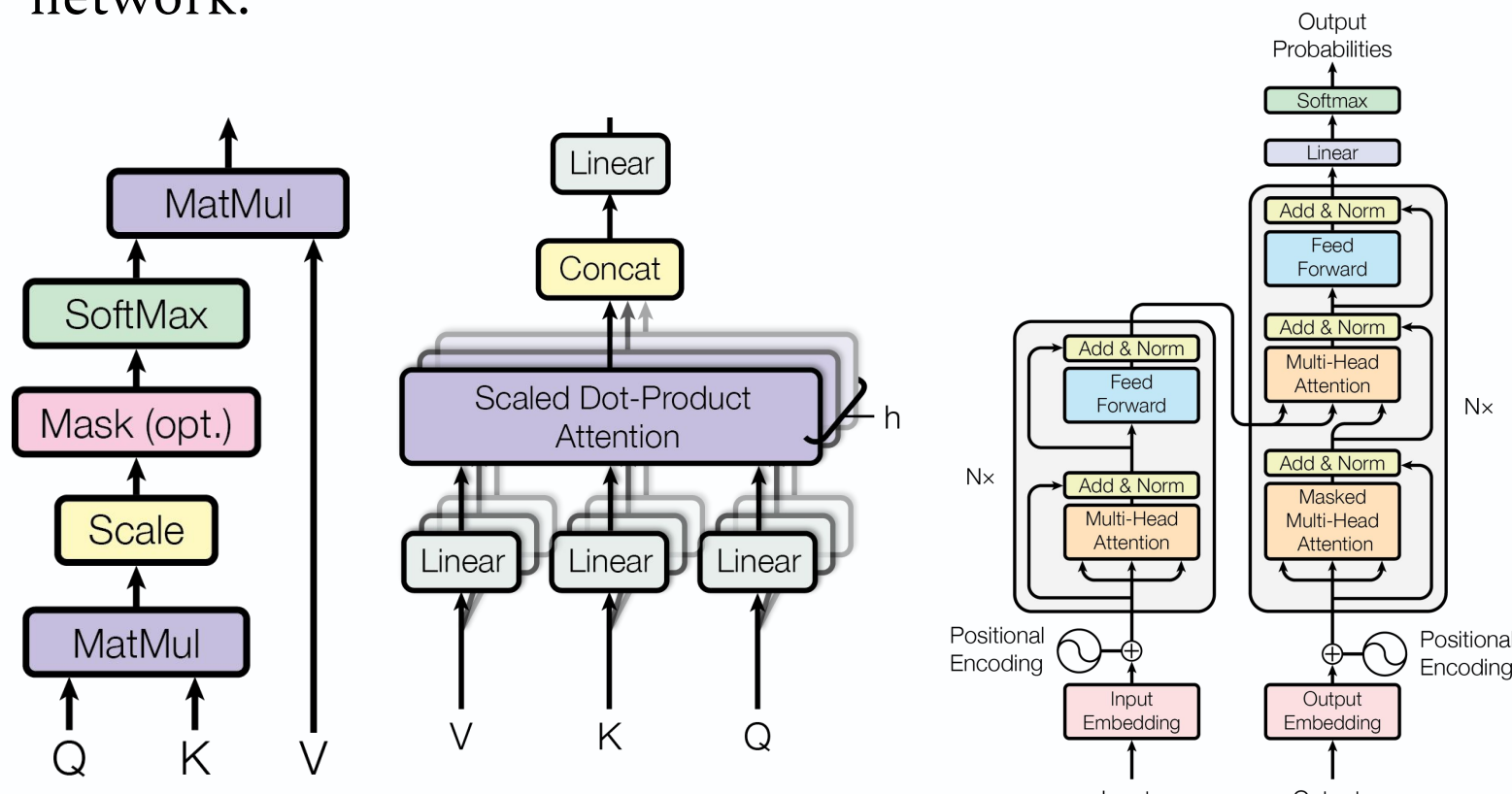


Fig. 1: (left) Scaled dot-product attention.  
(right) Multi head attention

Fig. 2: Transformer architecture

### 2.1. Positional encoding

Sinusoidal (fixed):

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{model}})$$
$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{model}})$$

Learned embeddings:

$$PE = \text{Embedding}(pos)$$

### 2.2. Layer norm

Post layer norm (original paper):  $\text{LayerNorm}(x + \text{Sublayer}(x))$

Pre layer norm (modern standart):  $x + \text{Sublayer}(\text{LayerNorm}(x))$

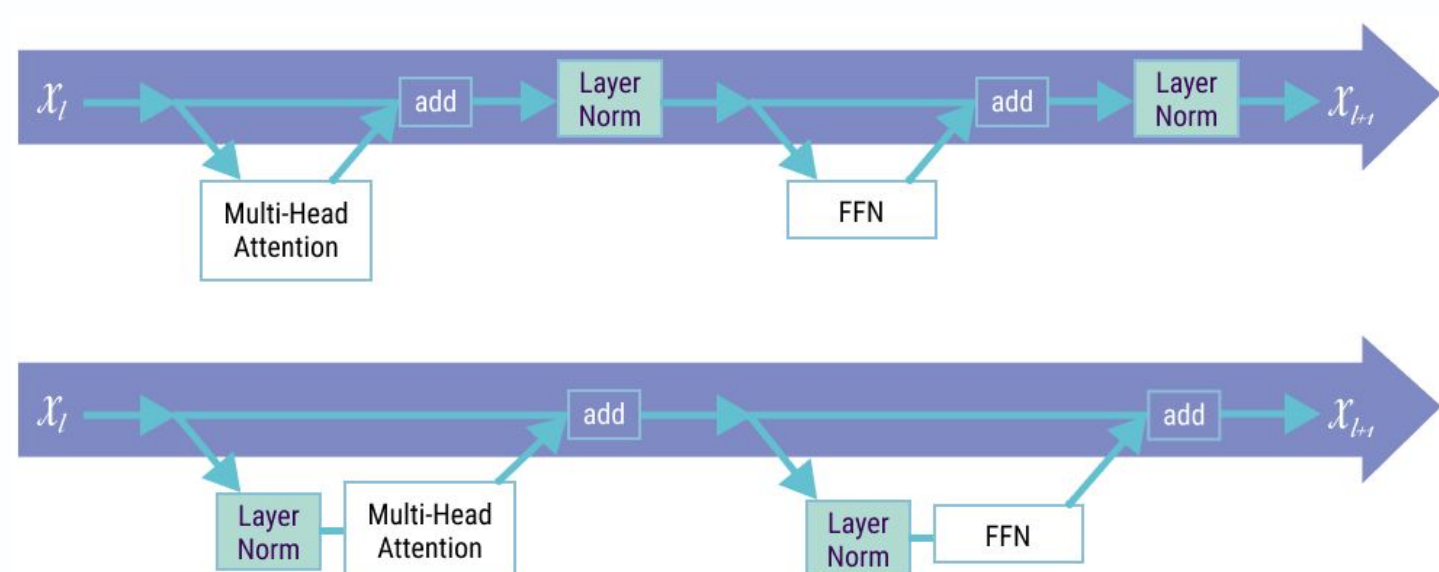


Fig. 3: (up) Post layer norm. (down) Pre layer norm

### 2.3. Activation functions

We examine common activation function – ReLU, GELU, Swish, GLU, and functions proposed by Shazeer (2020):

$$\text{FFN}_{\text{ReLU}}(x, W, V, W_2) = (\max(0, xW) \odot xV)W_2$$
$$\text{FFN}_{\text{GELU}}(x, W, V, W_2) = (\text{GELU}(xW) \odot xV)W_2$$
$$\text{FFN}_{\text{SwiGLU}}(x, W, V, W_2) = (\text{Swish}(xW) \odot xV)W_2$$

## 3. Datasets

We conduct experiments on the following datasets:

- Encoder-Decoder
  - Eng-Rus dataset — 790k pairs of sentences
- Decoder-only
  - Names dataset — 32k different names
  - Poetry dataset — 17k Russian classic poems

## 4. Experiments

### 4.1. Text generation

#### Models setup

**Names:** emb=8, ff\_n\_hidden=64, heads=8×64, seq\_len=10, dropout=0.3, batch=32. 100k parameters.

**Poetry:** emb=256, ff\_n\_hidden=384, heads=8×32, seq\_len=512, dropout=0.5, batch=128. 7 million parameters.

| Configuration                 | Train Loss | Val Loss | Train PPL | Val PPL | Best Val Loss | Best Val PPL |
|-------------------------------|------------|----------|-----------|---------|---------------|--------------|
| Post-LN + ReLU                | 2.2356     | 2.2233   | 9.3516    | 9.2375  | 2.2226        | 9.2312       |
| Post-LN + GELU                | 2.2349     | 2.2232   | 9.3456    | 9.2372  | 2.2232        | 9.2372       |
| Post-LN + GLU                 | 2.2368     | 2.2274   | 9.3631    | 9.2760  | 2.2274        | 9.2760       |
| Pre-LN + ReLU                 | 2.2576     | 2.2305   | 9.5603    | 9.3043  | 2.2305        | 9.3043       |
| Pre-LN + GLU                  | 2.2753     | 2.2433   | 9.7311    | 9.4244  | 2.2433        | 9.4244       |
| Post-LN + ReLU + sin/cos      | 2.2285     | 2.2166   | 9.2858    | 9.1763  | 2.2166        | 9.1763       |
| Post-LN + ReLU + learned      | 2.2067     | 2.2034   | 9.0857    | 9.0554  | 2.2028        | 9.0504       |
| Pre-LN + ReLU + sin/cos       | 2.2183     | 2.2008   | 9.1919    | 9.0318  | 2.2008        | 9.0318       |
| Pre-LN + ReLU + learned       | 2.2671     | 2.2364   | 9.6515    | 9.3593  | 2.2352        | 9.3481       |
| Pre-LN + GELU + sin/cos       | 2.2318     | 2.2131   | 9.3163    | 9.1444  | 2.2131        | 9.1444       |
| Pre-LN + SWISH + sin/cos      | 2.2349     | 2.2137   | 9.3459    | 9.1494  | 2.2137        | 9.1494       |
| Pre-LN + GLU + sin/cos        | 2.2737     | 2.2404   | 9.7158    | 9.3973  | 2.2404        | 9.3973       |
| Pre-LN + GLU + learned        | 2.2762     | 2.2365   | 9.7396    | 9.3609  | 2.2319        | 9.3178       |
| Pre-LN + FFN-GLU + sin/cos    | 2.2186     | 2.2110   | 9.1947    | 9.1246  | 2.2110        | 9.1246       |
| Pre-LN + FFN-ReGLU + sin/cos  | 2.2075     | 2.2048   | 9.0927    | 9.0684  | 2.2048        | 9.0684       |
| Pre-LN + FFN-GEGLU + sin/cos  | 2.2019     | 2.2044   | 9.0419    | 9.0652  | 2.2016        | 9.0395       |
| Pre-LN + FFN-SwiGLU + sin/cos | 2.2115     | 2.2108   | 9.1295    | 9.1233  | 2.2097        | 9.1129       |

Fig. 4: Results on names dataset

| Configuration                 | Train Loss | Val Loss | Train PPL | Val PPL  | Best Val Loss | Best Val PPL |
|-------------------------------|------------|----------|-----------|----------|---------------|--------------|
| Post-LN + ReLU + sin/cos      | 4.0291     | 4.4271   | 56.2120   | 83.6913  | 4.4271        | 83.6913      |
| Pre-LN + ReLU + sin/cos       | 4.6207     | 4.6835   | 101.5652  | 108.1518 | 4.6835        | 108.1518     |
| Pre-LN + GELU + sin/cos       | 4.5897     | 4.6620   | 98.4641   | 105.8477 | 4.6620        | 105.8477     |
| Pre-LN + SWISH + sin/cos      | 4.5920     | 4.6606   | 98.6891   | 105.7046 | 4.6606        | 105.7046     |
| Pre-LN + GLU + sin/cos        | 4.6462     | 4.7037   | 104.1915  | 110.3502 | 4.7037        | 110.3502     |
| Pre-LN + FFN-ReGLU + sin/cos  | 4.4937     | 4.7670   | 89.4526   | 117.5661 | 4.7644        | 117.2605     |
| Pre-LN + FFN-GEGLU + sin/cos  | 4.4978     | 4.7755   | 89.8192   | 118.5695 | 4.7755        | 118.5695     |
| Pre-LN + FFN-SwiGLU + sin/cos | 4.4327     | 4.7171   | 84.1574   | 111.8394 | 4.7171        | 111.8394     |

Fig. 5: Results on poetry dataset

#### Names

generation  
example:

semlyn, lailana,  
cheongua  
shesly, ebfemon,  
denallea  
deyr, onsist, hamaiel,  
motes

#### Poetry

generation  
example:

И в звон зубов лед и гроз,  
И мухомзы и озноб  
Зорь, чуть ослепленный  
завзвон.  
Лежит в печчитаньи шваль.  
Возванит, то летит роман,  
Весь высохший саблею  
блеснет,  
И в зелени

### 4.2. Machine translation

The ED architecture demonstrates 22-62% higher machine translation metrics than D-only.

| Metrics | ED     | D-Only | Diff    |
|---------|--------|--------|---------|
| BLEU    | 39.42  | 24.32  | ↓ 38.3% |
| ChrF++  | 60.72  | 43.56  | ↓ 28.3% |
| COMET   | 0.8308 | 0.682  | ↓ 17.9% |

#### Example:

**Source:**  
I didn't go to French  
class yesterday.

**Reference:**  
Я вчера не ходил на  
урок французского.

**ED output:**  
Я вчера не ходил  
на французский .

**D-only output:**  
Я вчера ходил на  
урок французского .

The model used a  
natural conversational  
simplification.

Word "не" is omitted,  
the meaning has  
changed to the opposite.

## 5. Conclusion

We implemented Transformer, compared ED vs D-only architectures, and tested various design choices. Findings: (1) ED outperforms D-only on translation; (2) GLU variants from Shazeer (2020) did not provide significant improvements over standard activations in D-only variant.